

Case Study 1 — Bayesian Linear Regression: Gradient-Based Optimisation and Posterior Variance Collapse

David Ewing (82171165)

11 May 2026

Abstract

Mean-field Variational Inference (VI) is useful because it turns Bayesian posterior computation into an optimisation problem, but this convenience comes with a cost. The approximation can become too confident, giving posterior variances that are smaller than they should be. In this case study, I examine that issue—posterior variance collapse—in a controlled Bayesian linear regression setting.

The linear case is chosen deliberately. It is simple enough for the main quantities to be checked directly, but still rich enough to show the difference between optimisation success and reliable Bayesian uncertainty. With Gaussian–Gamma conjugate priors, Coordinate Ascent Variational Inference (CAVI) gives closed-form updates and provides the analytical baseline for the numerical work. Gradient ascent, Newton’s method, and BFGS are then applied to the same ELBO so that their behaviour can be compared against that baseline.

The ELBO is derived under the mean-field factorisation

$$q(\boldsymbol{\beta}, \tau_e) = q(\boldsymbol{\beta})q(\tau_e),$$

where $q(\boldsymbol{\beta})$ is Gaussian and $q(\tau_e)$ is Gamma. The variational parameters are the coefficient mean $\boldsymbol{\mu}_\beta$, coefficient covariance $\boldsymbol{\Sigma}_\beta$, and Gamma parameters a_e and b_e . Because these quantities must remain valid during optimisation, the covariance matrix is represented through a Cholesky factor and the positive Gamma parameters are represented in log-space.

Each optimiser is described using the same course framework: starting point, search direction, step-size rule, and stopping condition. For the line-search methods, a proposed direction must be an ascent direction,

$$\nabla \text{ELBO}(\boldsymbol{\phi}^{(t)})^T \mathbf{d}^{(t)} > 0.$$

Gradient ascent uses Armijo backtracking, while BFGS uses Wolfe conditions to preserve the curvature condition needed for the inverse-Hessian update. Newton’s method is included to test whether curvature information improves the optimisation, while also exposing the practical difficulty that the Hessian may not always give a safe ascent direction.

The comparison is not limited to ELBO convergence. The methods are also compared on runtime, iteration count, posterior mean accuracy, and posterior standard-deviation ratios against the analytical posterior and a Gibbs sampler reference. A secondary experiment adds entropy regularisation to the ELBO to test whether explicitly rewarding broader covariance estimates can reduce posterior variance collapse.

The central question is therefore not only which method maximises the ELBO. It is also whether the resulting approximation remains honest about uncertainty. This distinction is the main reason for using the linear model first: the posterior, CAVI baseline, and Gibbs reference can all be checked in the same setting before moving to less tractable models.

1 Introduction

Bayesian linear regression is the simplest non-trivial model in which all components of variational inference—the ELBO, its gradient, and the posterior approximation—can be derived and verified in closed form, making it the natural baseline for comparing optimisation methods.

Bayesian inference provides principled uncertainty quantification but requires computing intractable posterior distributions. For most real-world models, the posterior integral,

$$p(\boldsymbol{\theta} \mid \mathbf{y}) = \frac{p(\mathbf{y} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta})}{\int p(\mathbf{y} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta}) d\boldsymbol{\theta}},$$

is typically intractable. In this case study, $\boldsymbol{\theta} = (\boldsymbol{\beta}, \tau_e)$ —the regression coefficients and the observation noise precision—so the posterior of interest is

$$p(\boldsymbol{\beta}, \tau_e \mid \mathbf{y}) \propto p(\mathbf{y} \mid \boldsymbol{\beta}, \tau_e) p(\boldsymbol{\beta}) p(\tau_e).$$

The substitution $\boldsymbol{\theta} = (\boldsymbol{\beta}, \tau_e)$ is specific to this case study; other case studies in this project define $\boldsymbol{\theta}$ according to their own model parameters. Conjugate Gaussian–Gamma priors make this case study a deliberate exception: the posterior is analytically tractable and CAVI coordinate updates are closed-form, providing a verified reference against which gradient-based VI methods can be compared.

Variational Inference (VI) converts this inference problem into an optimisation problem, making it directly amenable to the gradient-based methods in this course. Instead of computing the true posterior, VI finds the “best” approximation $q(\boldsymbol{\theta})$ from a tractable family $Q(\boldsymbol{\theta})$ by maximising the Evidence Lower Bound (ELBO). This transforms intractable integration into tractable optimisation—a natural testbed for the gradient-based methods.

A well-known failure mode of mean-field VI is *posterior variance collapse*: the optimised approximation $q(\boldsymbol{\theta})$ is systematically overconfident, underestimating posterior uncertainty. This is a direct consequence of the KL direction minimised during ELBO maximisation, compounded by the mean-field factorisation assumption.¹

I apply four optimisation methods—CAVI, gradient ascent, Newton’s method, and BFGS—to the ELBO and compare their convergence behaviour, computational cost, and posterior quality. As a secondary investigation, the ELBO is augmented with an entropy regularisation term $\lambda H[q(\boldsymbol{\beta})]$,

$$\mathcal{L}_{\text{reg}}(\boldsymbol{\phi}) = \mathcal{L}(\boldsymbol{\phi}) + \lambda H[q(\boldsymbol{\beta})],$$

to examine whether explicitly incentivising broader posteriors can mitigate variance collapse. Throughout all gradient-based methods, the variational covariance $\boldsymbol{\Sigma}_{\boldsymbol{\beta}}$ must remain symmetric positive definite at every iteration.²

The classical foundation for the variational framework used here is least-squares regression, as developed in lectures by Prof. Roy S. Smith (Erskine Visiting Fellow, ETH Zürich, ENEL445 2026). The design matrix $\mathbf{X}$ ³ and coefficient vector $\boldsymbol{\beta}$ ⁴ follow the standard over-determined formulation ($n > p$ ⁵), so that the normal equations $(\mathbf{X}^T \mathbf{X})\boldsymbol{\beta} = \mathbf{X}^T \mathbf{y}$ have a unique solution and the Bayesian posterior mean reduces to this solution in the limit of a flat prior. The noise model adopted here—errors on $\mathbf{y}$ only, with noise precision τ_e —corresponds to the equation-error form; this is an explicit modelling assumption rather than a derived result.

¹See ?? for the full mechanistic explanation.

²This structural constraint is satisfied automatically via Cholesky reparameterisation; see ??.

³Denoted Φ and called the *regressor matrix* in lectures (Prof. R. S. Smith, ENEL445 2026).

⁴Denoted θ in lectures.

⁵Denoted d in lectures.

1.1 Numerical Study and Required Optimisation Components

I formulate variational inference for Bayesian linear regression as a constrained optimisation problem and apply four methods: Coordinate Ascent Variational Inference (CAVI), gradient ascent, Newton's method, and BFGS. I characterise each method using the same four components from the course framework: entry conditions, search direction, line search, and exit conditions. All four methods share the same initial point so that comparisons reflect method behaviour rather than initialisation.

Entry Conditions

All methods are initialised at the prior mean: $\boldsymbol{\mu}_\beta = \mathbf{0}$, $\boldsymbol{\Sigma}_\beta = \mathbf{I}_p$, $a_e = \alpha_e$, $b_e = \gamma_e$. In the unconstrained parameterisation this corresponds to $\mathbf{L} = \mathbf{I}_p$ (so $\tilde{\ell}_{ii} = 0$ and off-diagonal entries zero), $\eta_a = \log \alpha_e$, and $\eta_b = \log \gamma_e$. CAVI does not require an explicit initialisation of the search direction and uses the same starting variational parameters.

Search Direction

For all gradient-based methods, the proposed search direction must be an ascent direction before a step is accepted:

$$\nabla \text{ELBO}(\boldsymbol{\phi}^{(t)})^T \mathbf{d}^{(t)} > 0. \quad (1)$$

Gradient ascent uses $\mathbf{d}^{(t)} = \nabla \mathcal{L}(\boldsymbol{\xi}^{(t)})$ directly. Newton's method uses $\mathbf{d}^{(t)} = -[\nabla^2 \mathcal{L}(\boldsymbol{\xi}^{(t)})]^{-1} \nabla \mathcal{L}(\boldsymbol{\xi}^{(t)})$, falling back to the gradient direction if the Hessian is not positive definite. BFGS approximates the inverse Hessian as $\mathbf{B}_t$ and uses $\mathbf{d}^{(t)} = \mathbf{B}_t \nabla \mathcal{L}(\boldsymbol{\xi}^{(t)})$. CAVI replaces the search direction with closed-form coordinate updates derived analytically from the model.

Line Search

The step size $\alpha^{(t)}$ is determined per method. Gradient ascent uses Armijo backtracking: starting from an initial step size, $\alpha^{(t)}$ is halved until sufficient increase in the ELBO is achieved. BFGS uses Wolfe conditions, combining sufficient increase with a curvature condition that ensures

$$\mathbf{y}_t^T \mathbf{s}_t > 0, \quad (2)$$

where $\mathbf{s}_t = \boldsymbol{\xi}^{(t+1)} - \boldsymbol{\xi}^{(t)}$ and $\mathbf{y}_t = \nabla \mathcal{L}(\boldsymbol{\xi}^{(t+1)}) - \nabla \mathcal{L}(\boldsymbol{\xi}^{(t)})$. This positivity condition is required for the BFGS update to preserve positive definiteness of $\mathbf{B}_t$. Newton's method uses a fixed step size of 1 unless the Hessian is indefinite. CAVI has no line search: each coordinate update exactly maximises the ELBO with respect to that parameter in closed form.

Exit Conditions

Exit conditions are applied uniformly across the optimisation methods. Iteration terminates when any of the following conditions is satisfied:

- (i) Gradient norm: $\|\nabla \text{ELBO}\| < \varepsilon_g$.
- (ii) Relative ELBO change: $|\Delta \text{ELBO}|/|\text{ELBO}| < \varepsilon_r$.
- (iii) Parameter change: $\|\Delta \boldsymbol{\phi}\| < \varepsilon_\phi$.
- (iv) Maximum iterations: $t \geq t_{\max}$.

CAVI additionally monitors the maximum relative change in its variational parameters:

$$\max_i \frac{|\phi_i^{(t+1)} - \phi_i^{(t)}|}{|\phi_i^{(t+1)}|} < \varepsilon. \quad (3)$$

1.2 Bayesian Regression Model

The Bayesian regression model is

$$\mathbf{y} \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \tau_e^{-1}\mathbf{I}_n) \quad (\text{likelihood}), \quad (4)$$

$$\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \lambda_\beta^{-1}\mathbf{I}_p) \quad (\text{prior on coefficients}), \quad (5)$$

$$\tau_e \sim \text{Gamma}(\alpha_e, \gamma_e) \quad (\text{prior on precision}). \quad (6)$$

Here, $\mathbf{X}$ is the design matrix, $\mathbf{y}$ is the response vector, $\boldsymbol{\beta}$ is the coefficient vector, and τ_e is the observation-noise precision.

1.3 Mean-Field Variational Approximation

The mean-field approximation is

$$q(\boldsymbol{\beta}, \tau_e) = q(\boldsymbol{\beta}) q(\tau_e), \quad (7)$$

$$q(\boldsymbol{\beta}) = \mathcal{N}(\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta), \quad (8)$$

$$q(\tau_e) = \text{Gamma}(a_e, b_e). \quad (9)$$

The variational parameters are

$$\boldsymbol{\phi} = (\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta, a_e, b_e), \quad (10)$$

with dimension $d = p + p(p+1)/2 + 2$. For $p = 3$, this gives $d = 3 + 6 + 2 = 11$ parameters to optimise.

1.4 Key Notation

Symbol	Meaning
Bayesian model	
$\mathbf{X}$	Design matrix ($n \times p$)
$\mathbf{y}$	Response vector
$\boldsymbol{\beta}$	Regression coefficient vector
τ_e	Observation-noise precision
α_e, γ_e	Shape and rate of the prior on τ_e
p	Number of predictors
n	Number of observations
Variational approximation	
$q(\cdot)$	Variational approximating distribution
$\boldsymbol{\phi}$	Full variational parameter vector $(\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta, a_e, b_e)$
$\boldsymbol{\mu}_\beta$	Mean of variational posterior $q(\boldsymbol{\beta})$
$\boldsymbol{\Sigma}_\beta$	Covariance of variational posterior $q(\boldsymbol{\beta})$
a_e, b_e	Shape and rate of variational $q(\tau_e) = \text{Gamma}(a_e, b_e)$
$\mathcal{L}(\boldsymbol{\phi})$	Evidence Lower Bound (ELBO)
$H[q(\boldsymbol{\beta})]$	Differential entropy of $q(\boldsymbol{\beta})$
λ	Entropy regularisation weight
Reparameterisation	
$\mathbf{L}$	Lower-triangular Cholesky factor, $\boldsymbol{\Sigma}_\beta = \mathbf{L}\mathbf{L}^T$
$\tilde{\ell}_{ii}$	Log-diagonal entry, $L_{ii} = e^{\tilde{\ell}_{ii}}$ (ensures $L_{ii} > 0$)
$\boldsymbol{\xi}$	Unconstrained optimisation vector $(\boldsymbol{\mu}_\beta, \tilde{\ell}_{11}, \ell_{21}, \dots, \eta_a, \eta_b)$
η_a, η_b	Log-space parameters, $a_e = e^{\eta_a}$, $b_e = e^{\eta_b}$
Optimisation	
$\mathbf{d}^{(t)}$	Search direction at iteration t
α_t	Step size (line-search parameter) at iteration t
$\nabla \mathcal{L}$	Gradient of the ELBO w.r.t. $\boldsymbol{\xi}$
$\nabla^2 \mathcal{L}$	Hessian of the ELBO w.r.t. $\boldsymbol{\xi}$
$\mathbf{B}_t$	BFGS inverse-Hessian approximation at iteration t
Special functions	
$\psi(x)$	Digamma function, $\psi(x) = \frac{d}{dx} \log \Gamma(x)$
$\psi^{(1)}(x)$	Trigamma function, $\psi^{(1)}(x) = \frac{d}{dx} \psi(x)$
Lecture correspondence (Prof. R. S. Smith, ENEL445 2026)	
Φ	Regressor matrix = $\mathbf{X}$ in this report
θ	Coefficient vector = $\boldsymbol{\beta}$ in this report
d	Number of parameters = p in this report
$J(\theta)$	Cost function = $\mathcal{L}(\boldsymbol{\phi})$ (ELBO) in this report

2 Optimisation Problem Formulation

All four methods—CAVI, gradient ascent, Newton’s method, and BFGS—operate on the same objective and respect the same constraints. The formulation below follows the standard form from the course, so that differences in performance reflect the methods themselves rather than problem differences.

Decision Variables

For Bayesian linear regression with p predictors, the variational parameters (decision variables) are:

- $\boldsymbol{\mu}_\beta \in \mathbb{R}^p$: posterior mean of regression coefficients
- $\boldsymbol{\Sigma}_\beta \in \mathbb{R}^{p \times p}$: posterior covariance (symmetric positive definite, $\boldsymbol{\Sigma}_\beta \succ 0$)
- $a_e, b_e > 0$: shape and rate parameters of the variational Gamma factor for precision

Here, a_e and b_e parameterise $q(\tau_e) = \text{Gamma}(a_e, b_e)$; the model prior on τ_e is specified in the Test Case section.

Total dimension: $d = p + \frac{p(p+1)}{2} + 2 = 11$ variables for $p = 3$ predictors.

Objective Function

To introduce the objective, I start from two compact identities:

$$\text{KL}(q||p) = \underbrace{-H[q]}_{\text{negative entropy of } q} - \underbrace{\mathbb{E}_q[\log p(\boldsymbol{\beta})]}_{\text{expected log-prior}}, \quad (11)$$

$$\mathcal{L}(\theta) = \underbrace{\mathbb{E}_q[\log p(\mathbf{y} | \boldsymbol{\beta})]}_{\text{fit the data}} - \underbrace{\text{KL}(q||p)}_{\text{stay close to prior}}. \quad (12)$$

Step 1: the expected log-likelihood term rewards explanations of the observed data.

Step 2: the KL term prevents overfitting by penalising departure from the prior structure.

Step 3: in this report, these two ideas are written in joint-form ELBO notation over $(\boldsymbol{\beta}, \tau_e)$, which is the objective optimised by all methods.

Maximise the Evidence Lower Bound (ELBO):

$$\text{ELBO}(\phi) = \mathbb{E}_q[\log p(\mathbf{y}, \boldsymbol{\beta}, \tau_e)] + H(q) \quad (13)$$

$$= \mathbb{E}_q[\log p(\mathbf{y}, \boldsymbol{\beta}, \tau_e)] - \mathbb{E}_q[\log q(\boldsymbol{\beta}, \tau_e)]. \quad (14)$$

where $\phi = (\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta, a_e, b_e)$ are the variational parameters. The first term, $\mathbb{E}_q[\log p(\mathbf{y}, \boldsymbol{\beta}, \tau_e)]$, is the expected log joint: it rewards approximations q that place mass where the data and prior are jointly well explained. The second term, $H(q) = -\mathbb{E}_q[\log q(\boldsymbol{\beta}, \tau_e)]$, is the *differential entropy* of the approximating distribution: it rewards approximations that are broad and diffuse, penalising overconfident solutions. Maximising the ELBO therefore balances data fit against posterior spread — the entropy term is precisely what variance collapse works against.

Constraints

- $\boldsymbol{\Sigma}_\beta \succ 0$: Covariance must be strictly positive definite; $\boldsymbol{\Sigma}_\beta^{-1}$ appears in the ELBO and CAVI updates, so a semi-definite covariance causes those computations to fail.
- $a_e, b_e > 0$: Gamma parameters must be positive
- Optional: Minimum variance constraints $\sigma_{\beta_j}^2 \geq \sigma_{\min}^2$ to prevent underdispersion

Constraint Handling via Reparameterisation:

To enforce $\Sigma_\beta \succ 0$, use Cholesky decomposition:

$$\Sigma_\beta = \mathbf{L}\mathbf{L}^T \quad (15)$$

where $\mathbf{L} \in \mathbb{R}^{p \times p}$ is lower triangular with positive diagonal entries. The unconstrained parameters are:

- $L_{ii} = e^{\ell_{ii}}$ for $i = 1, \dots, p$ (exponentiate to ensure positivity)
- $L_{ij} = \ell_{ij}$ for $i > j$ (off-diagonal entries unconstrained)

Total Cholesky parameters: $\frac{p(p+1)}{2}$ elements of ℓ .

To enforce $a_e, b_e > 0$, use log-space parameterisation:

$$a_e = e^{\eta_a}, \quad b_e = e^{\eta_b} \quad (16)$$

where $\eta_a, \eta_b \in \mathbb{R}$ are unconstrained.

Unconstrained parameter vector: $\xi = (\mu_\beta, \ell, \eta_a, \eta_b) \in \mathbb{R}^d$ with $d = p + \frac{p(p+1)}{2} + 2 = 11$ for $p = 3$.

3 Posterior Variance Collapse

Because this case study uses conjugate Gaussian–Gamma priors, the true posterior $p(\beta, \tau_e \mid \mathbf{y})$ is available as a Normal–Gamma distribution. That makes the linear case a useful check on the whole numerical study. Posterior variance collapse can be measured against an exact reference, rather than inferred only from a sampler. The Gibbs sampler is still included, but here it is a practical comparison point rather than the only standard of comparison.

In this report, posterior variance collapse means that the variational covariance Σ_β is too small. The most visible symptom is that the diagonal entries of Σ_β understate the uncertainty in the regression coefficients. The posterior intervals then look more precise than the evidence justifies. For that reason, the results are read in two ways. First, the optimisers are judged in the usual numerical sense: whether they reach a high ELBO, how quickly they converge, and how much computation they require. Second, the resulting posterior spread is checked directly. A good optimiser is not automatically a good uncertainty approximation.

3.1 Why Collapse Occurs

The first source of collapse is the mean-field assumption itself:

$$q(\beta, \tau_e) = q(\beta) q(\tau_e). \quad (17)$$

This assumption is convenient, but it removes the posterior dependence between the regression coefficients and the noise precision. In the exact Normal–Gamma posterior, a larger noise precision is associated with tighter coefficient estimates. Once the approximation forces β and τ_e to be independent, part of that joint uncertainty can no longer be represented.

The second cause is the direction of the KL divergence implied by the ELBO. Standard VI minimises

$$\begin{aligned} \text{KL}[q \parallel p] &= \int q(\theta) \log \frac{q(\theta)}{p(\theta \mid \mathbf{y})} d\theta \\ &= \int_0^\infty \int_{\mathbb{R}^p} q(\beta, \tau_e) \log \frac{q(\beta, \tau_e)}{p(\beta, \tau_e \mid \mathbf{y})} d\beta d\tau_e \\ &= \int_0^\infty \int_{\mathbb{R}^p} q(\beta) q(\tau_e) \log \frac{q(\beta) q(\tau_e)}{p(\beta, \tau_e \mid \mathbf{y})} d\beta d\tau_e. \end{aligned} \quad (18)$$

The three-line integral is deceptively simple to implement. Because each variational factor belongs to the exponential family, both KL contributions reduce to closed-form entropy formulas (see Appendix ??), so the entire KL computation reduces to two lines in `linear_run.ipynb`:

```
# -E[log q(beta)] = Gaussian entropy (closed-form)
sign, logdet = np.linalg.slogdet(Sigma)
neg_lq_beta = 0.5 * logdet + 0.5 * p * (1 + np.log(2 * np.pi))

# -E[log q(tau_e)] = Gamma entropy (closed-form)
neg_lq_tau = a_e - np.log(b_e) + gammaln(a_e) + (1 - a_e) * digamma(a_e)
```

This objective penalises q heavily when it puts probability mass in regions where the true posterior is small. It is less severe when q simply fails to cover parts of the true posterior support. In practice, the approximation is encouraged to sit on the high-density centre of $p(\boldsymbol{\beta}, \tau_e | \mathbf{y})$ rather than cover the full posterior spread. Since the analytical posterior is available in this case, this narrowing can be measured directly.

3.2 Diagnostic

The numerical experiments compare the variational posterior against two references: the analytically derived true posterior (available because of conjugate priors) and a Gibbs sampler. For each coefficient β_j , two standard-deviation ratios are computed:

$$\frac{\sigma_{\text{VB}}}{\sigma_{\text{true}}}, \quad \frac{\sigma_{\text{VB}}}{\sigma_{\text{Gibbs}}}. \quad (19)$$

Ratios below one confirm that the variational approximation is narrower than the reference (*posterior variance collapse*). The ratio against the true posterior is the definitive diagnostic; the ratio against Gibbs serves as a consistency check and provides the comparison metric used in the other case studies where the true posterior is not available.

3.3 Approaches Considered

Three approaches are considered for reducing or diagnosing posterior variance collapse. Of these, only entropy regularisation (Approach 3) is implemented in this report; the first two are discussed to motivate that choice.

3.3.1 Informative Prior on τ_e

A stronger prior on τ_e can limit how large the expected precision becomes. Since larger precision corresponds to smaller observation-noise variance, controlling τ_e can indirectly prevent $\boldsymbol{\Sigma}_\beta$ from becoming too small. The limitation is that this changes the statistical model and requires a defensible prior choice.

3.3.2 Minimum Variance Constraint

A direct constraint can be imposed on the diagonal entries of $\boldsymbol{\Sigma}_\beta$:

$$\Sigma_{\beta,jj} \geq \sigma_{\min}^2, \quad j = 1, \dots, p. \quad (20)$$

This prevents individual coefficient variances from falling below a chosen threshold. The limitation is that the threshold is external to the original ELBO objective.

3.3.3 Entropy Regularisation

The most direct objective-based approach considered here is entropy regularisation:

$$\mathcal{L}_{\text{reg}}(\phi) = \mathcal{L}(\phi) + \lambda H[q(\beta)]. \quad (21)$$

For $q(\beta) = \mathcal{N}(\mu_\beta, \Sigma_\beta)$, the entropy is

$$H[q(\beta)] = \frac{1}{2} \log \det(2\pi e \Sigma_\beta) \quad (22)$$

$$= \frac{p}{2}(1 + \log 2\pi) + \frac{1}{2} \log \det(\Sigma_\beta). \quad (23)$$

The regularised gradient with respect to the covariance matrix gains the additional term

$$\nabla_{\Sigma_\beta} \mathcal{L}_{\text{reg}} = \nabla_{\Sigma_\beta} \mathcal{L} + \frac{\lambda}{2} \Sigma_\beta^{-1}. \quad (24)$$

This term rewards broader covariance estimates and directly counters the tendency toward overly narrow posterior approximations.

4 ELBO Evaluation

Before comparing the optimisation methods, the ELBO implementation is checked numerically. This step is mainly a safeguard: if the gradient or reparameterisation is wrong, later convergence plots would be difficult to interpret.

4.1 Gradient Verification

The analytical gradient $\nabla_{\xi} \mathcal{L}$ is computed with respect to all unconstrained parameters $\xi = (\mu_\beta, \tilde{\ell}_{11}, \tilde{\ell}_{21}, \tilde{\ell}_{22}, \eta_a, \eta_b)$. Each component is checked against a central finite-difference approximation:

$$\frac{\partial \mathcal{L}}{\partial \xi_i} \approx \frac{\mathcal{L}(\xi + h e_i) - \mathcal{L}(\xi - h e_i)}{2h}, \quad h = 10^{-5}. \quad (25)$$

The relative error between the analytical and numerical gradient must satisfy $\|\nabla_{\text{analytical}} - \nabla_{\text{FD}}\|_{\infty} / \|\nabla_{\text{analytical}}\|_{\infty} \leq 10^{-5}$ for all components before any optimiser is run. The result is displayed as a bar chart (`linear_elbo_gradient_check.png`) comparing analytical and finite-difference values side by side for each unconstrained parameter. The gradient check table is saved to `results/tables/linear_gradient_check.tex`.

4.2 ELBO Landscape

A two-dimensional slice of the ELBO surface is computed by varying μ_{β_0} and μ_{β_1} over a grid while holding all other parameters fixed at their CAVI optimum. This visualises the objective surface that the gradient-based optimisers are navigating and confirms that the CAVI solution lies at the maximum. The result is saved as `linear_elbo_landscape.png`.

5 Optimisation Methods

The optimisation methods are used for more than finding a maximum of the ELBO. Each method also produces a variational posterior, and that posterior has to be checked for uncertainty collapse. The comparison therefore has both an optimisation side and an inference side.

5.1 CAVI Baseline

CAVI is used as the analytical baseline because the linear Gaussian–Gamma model is conjugate. Instead of choosing a generic search direction, CAVI updates one variational block at a time using closed-form expressions:

$$\Sigma_\beta \leftarrow \left(\frac{a_e}{b_e} \mathbf{X}^T \mathbf{X} + \lambda_\beta \mathbf{I}_p \right)^{-1}, \quad (26)$$

$$\boldsymbol{\mu}_\beta \leftarrow \frac{a_e}{b_e} \Sigma_\beta \mathbf{X}^T \mathbf{y}, \quad (27)$$

$$a_e \leftarrow \alpha_e + \frac{n}{2}, \quad (28)$$

$$b_e \leftarrow \gamma_e + \frac{1}{2} [\|\mathbf{y} - \mathbf{X} \boldsymbol{\mu}_\beta\|^2 + \text{tr}(\mathbf{X}^T \mathbf{X} \Sigma_\beta)]. \quad (29)$$

CAVI is expected to increase the ELBO monotonically⁶, with linear convergence and $O(np^2)$ cost per iteration. In this report it is the reference point, not the method being challenged in the conjugate case.

5.2 Gradient Ascent

Gradient ascent applies the first-order update in the unconstrained parameterisation:

$$\boldsymbol{\xi}^{(t+1)} = \boldsymbol{\xi}^{(t)} + \alpha_t \nabla \text{ELBO}(\boldsymbol{\xi}^{(t)}). \quad (30)$$

The search direction is the gradient, so the ascent condition is satisfied whenever the gradient is non-zero:

$$\nabla \text{ELBO}(\boldsymbol{\xi}^{(t)})^T \nabla \text{ELBO}(\boldsymbol{\xi}^{(t)}) > 0. \quad (31)$$

The step size is selected using Armijo backtracking so that each accepted step produces sufficient increase in the ELBO.

5.3 Newton’s Method

Newton’s method uses second-order curvature information. For maximisation, the ideal Newton direction is

$$\mathbf{d}^{(t)} = -[\nabla^2 \text{ELBO}(\boldsymbol{\xi}^{(t)})]^{-1} \nabla \text{ELBO}(\boldsymbol{\xi}^{(t)}). \quad (32)$$

Near a well-conditioned local maximum, Newton’s method can converge quadratically. Away from the optimum, however, the Hessian may fail to be negative definite, so the raw Newton direction may not be an ascent direction. Regularisation is therefore applied when necessary so that the search direction satisfies

$$\nabla \text{ELBO}(\boldsymbol{\xi}^{(t)})^T \mathbf{d}^{(t)} > 0. \quad (33)$$

5.4 BFGS

BFGS is included as the practical middle ground between gradient ascent and a full Newton method. It does not form the Hessian directly; instead, it builds inverse-curvature information from successive parameter and gradient changes. Define

$$\mathbf{s}_t = \boldsymbol{\xi}^{(t+1)} - \boldsymbol{\xi}^{(t)}, \quad (34)$$

$$\mathbf{y}_t = \nabla \text{ELBO}(\boldsymbol{\xi}^{(t+1)}) - \nabla \text{ELBO}(\boldsymbol{\xi}^{(t)}). \quad (35)$$

⁶Monotonic ELBO increase means the objective is non-decreasing across iterations: $\mathcal{L}^{(t+1)} \geq \mathcal{L}^{(t)}$. In CAVI, each coordinate update maximises the ELBO with respect to one variational factor while holding the others fixed, so each step cannot decrease the ELBO.

The BFGS update requires the curvature condition

$$\mathbf{y}_t^T \mathbf{s}_t > 0 \quad (36)$$

to maintain positive definiteness of the approximate inverse Hessian. Wolfe line-search conditions are used for this reason. BFGS has $O(\dim(\boldsymbol{\xi})^2)$ per-iteration cost and is expected to provide a practical balance between gradient ascent and Newton’s method.

5.5 Shared Ascent Check and Exit Conditions

For all line-search methods, the search direction must satisfy the ascent condition before a step is accepted:

$$\nabla \text{ELBO}(\boldsymbol{\xi}^{(t)})^T \mathbf{d}^{(t)} > 0. \quad (37)$$

Exit conditions are applied uniformly across methods using gradient norm, relative ELBO change, parameter change, and maximum iteration count. The detailed implementation of these conditions is given in Section 6.

6 Test Cases and Implementation Details

The numerical work uses one generated linear regression data set. The full implementation structure—ELBO evaluation, optimisers, and Gibbs sampler—is recorded in Appendix ??.

6.1 Test Case 1: Linear Bayesian Regression

The first test case is a linear Bayesian regression problem. It is the baseline setting because the data-generating process matches the model being fitted:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad (38)$$

$$\varepsilon_i \sim \mathcal{N}(0, \sigma_e^2). \quad (39)$$

The design matrix is

$$\mathbf{X} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}. \quad (40)$$

This case is used to check whether each optimisation method can reproduce the CAVI solution and to measure posterior variance collapse in the simplest regression setting.

```
def generate_linear_case(n=100, sigma=0.5, seed=1):
    rng = np.random.default_rng(seed)
    x = rng.normal(size=n)
    X = np.column_stack([np.ones(n), x])
    beta_true = np.array([1.0, 2.0])
    y = X @ beta_true + rng.normal(scale=sigma, size=n)
    return X, y, beta_true
```

6.2 Code Structure

The following workflow is applied to the linear test case:

1. Generate $\mathbf{X}$, $\mathbf{y}$, and β_{true} .
2. Fit all four methods (CAVI, gradient ascent, Newton's method, BFGS) and record the ELBO trajectory for each.
3. Run the Gibbs sampler reference implementation.
4. Compare posterior means, posterior standard deviations, convergence speed, and runtime.

```
X, y, beta_true = generate_linear_case()
results = run_all_methods(X, y, beta_true)
```

7 Assessment Framework

Each method is assessed against three criteria. The first is convergence: whether the method reaches the same ELBO value as the CAVI fixed point. The second is posterior accuracy: whether the variational posterior matches the Gibbs sampler and, in this linear case, the analytically available posterior. The Gibbs sampler is used as an uncertainty reference, not as an ELBO optimiser, so it is not judged on optimisation speed. The third is computational cost, measured using wall-clock time and iteration count. Analytical gradients are checked against central finite differences to relative error $\leq 10^{-5}$ before the optimiser comparison is made.

8 Results and Discussion

8.1 Test Case: Linear Bayesian Regression

The test case uses $n = 50$ observations generated from $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ with true parameters $\beta_0 = 2.0$, $\beta_1 = 1.5$, and noise precision $\tau_e = 4.0$ (noise standard deviation $\sigma_e = 0.5$). Figure 1 shows the generated dataset with the prior and likelihood overlays.

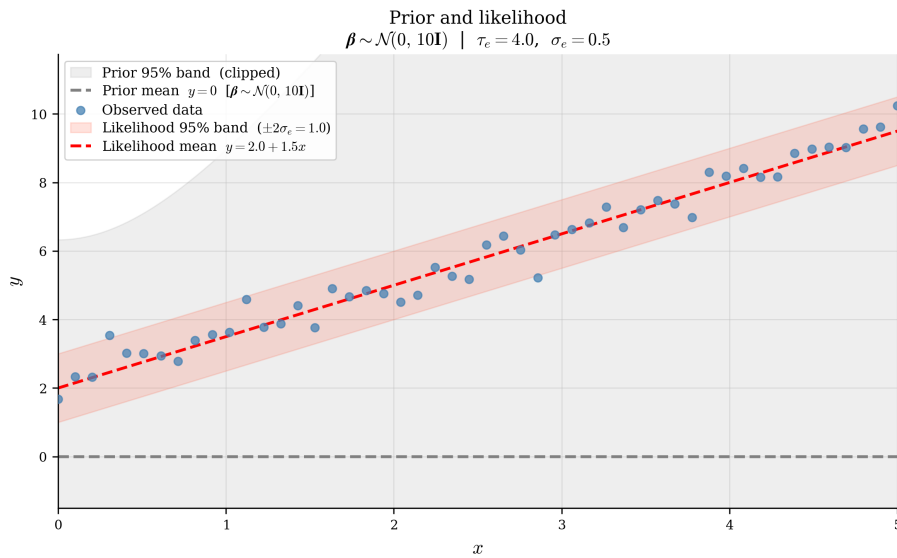


Figure 1: Prior, likelihood, and observed data for the linear model. The grey dashed line is the prior mean ($y = 0$) with a clipped grey 95% prior band, and the red dashed line is the likelihood mean ($y = 2.0 + 1.5x$) with its 95% band ($\pm 2\sigma_e = 1.0$).

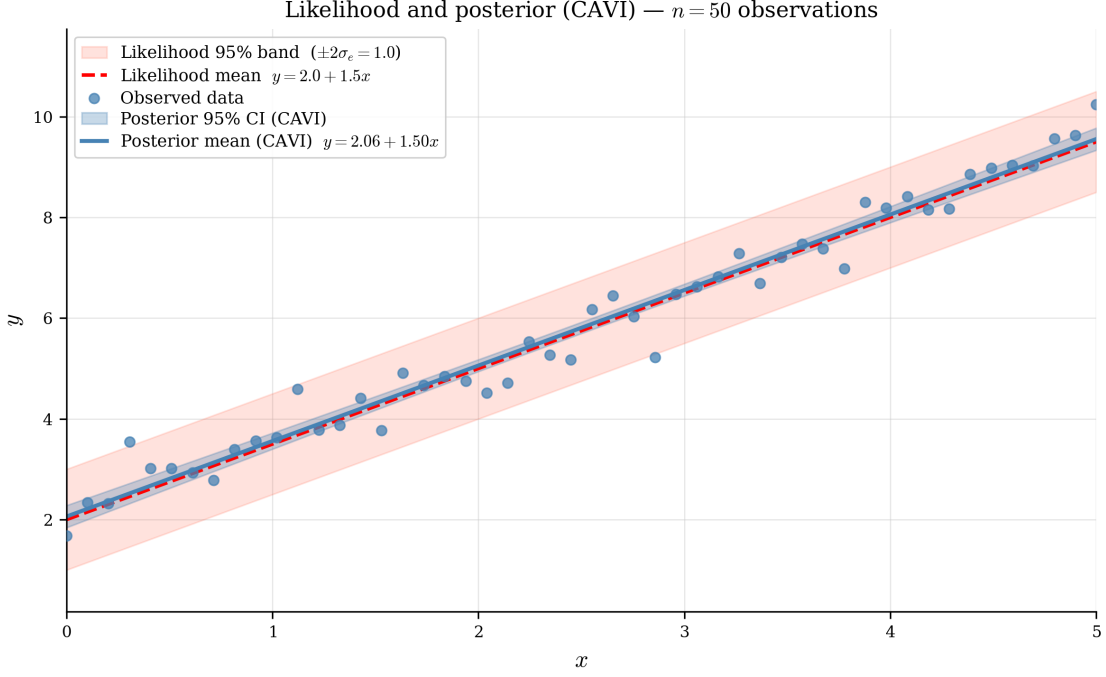


Figure 2: Likelihood and CAVI posterior for the linear predictor with $n = 50$ observations. The red dashed line and red band show the likelihood mean and 95% band, while the blue line and blue band show the CAVI posterior mean and 95% credible interval. The posterior sits close to the likelihood trend while providing a tighter uncertainty summary.

8.1.1 Posterior Contraction and the Role of Posterior Collapse

Posterior contraction (applicable here). Figure 2 illustrates a fundamental property of Bayesian inference: *posterior contraction*. Under the prior $\beta \sim \mathcal{N}(\mathbf{0}, 10\mathbf{I})$, the 95% predictive interval for the regression mean spans roughly $\pm 2\sqrt{10(1+x^2)} \approx \pm 6\text{--}20$ units across the data range, reflecting near-total ignorance about β before any data are seen. After conditioning on $n = 50$ observations, the CAVI posterior 95% credible interval for the regression mean shrinks to a few tenths of a unit — an order-of-magnitude reduction in uncertainty. This is exactly what Bayes’ theorem guarantees: as n increases, the likelihood dominates the prior and the posterior concentrates around the true parameter values at a rate of $O(n^{-1/2})$ per parameter [vaart1998asymptotic](#). The tight blue band in Figure 2 is therefore a feature, not a defect: it shows that 50 observations are sufficient to identify the linear relationship almost exactly. Outside this report context, however, that visual overlap could be misread as posterior collapse, so the figure should be interpreted together with the SD-ratio and Gibbs comparisons. Crucially, the remaining width of the blue band represents genuine, irreducible uncertainty about the *regression line*; the much wider tomato band shows the observation noise $\pm 2\sigma_e = \pm 1.0$, which is a separate quantity describing where individual data points fall, not where the mean lies.

Posterior collapse (not applicable in this case study). Posterior collapse is a distinct, pathological failure mode that can afflict mean-field variational inference in *hierarchical* models. It occurs when the CAVI updates for a precision hyperparameter τ_β drive that precision to infinity: the factor $q(\tau_\beta)$ concentrates on an implausibly large value, which in turn forces $q(\beta)$ to collapse toward the prior mean, discarding information from the data. Each update reinforces the other in a degenerate fixed point that maximises the ELBO locally but not globally [turner2011two](#). The collapse can be detected by comparing CAVI posterior variances against those of a Gibbs sampler: a CAVI variance dramatically smaller than Gibbs variance, without a corresponding gain in posterior mean accuracy, is a red flag.